2081:

Long Answer Questions: [10 marks each]

1. Why is it necessary to analyze the simulation output? Explain different estimation methods which are used in simulation output analysis.
2. Consider that a machine tool in a manufacturing shop is turning out parts at the rate of two every 5 minutes. As they are finished, the parts go to an inspector, who takes 5±2 minutes to examine each one and rejects about 15% of the parts. Now develop a block diagram and write the code for simulating the above problem using GPSS and also explain the function of each block used in the block diagram in detail.
3. Explain the independence and uniformity property of random number. For the following sample of random numbers, perform test for independence using K-S test. ($D_{0.05,10}$ = 0.41)

   0.35, 0.77, 0.12, 0.33, 0.88, 0.45, 0.19, 0.25, 0.91, 0.54

Short Answer Questions: [5 marks each]

4. Difference between static physical and dynamic physical models.
5. Describe different phases of simulation study with help of flowchart.
6. Explain Markov chain with suitable example. What are different application areas of Markov chain?
7. Explain iterative process of calibrating a simulation model.

8. Describe the process of model building, verification and validation in detail with example.
9. Define traffic intensity and server utilization. Write down the Kendall's notation for queuing system with example.
10. Define traffic intensity and server utilization. Write down the Kendall's notation for queuing system with example.
11. What is analog computer? Explain with suitable example.
12. Write short notes on:
    a. Queuing discipline
    b. Random variate

2080:

Long Answer Questions: [10 marks each]

1. Why is model of a system built? What is static model? Differentiate between static and dynamic mathematical models in simulation.
2. What is storage in GPSS? Describe the blocks associated to storage in GPSS. A machine tool in a manufacturing shop is turning out parts at the rate off one every 5 minutes. As they are finished the parts go on an inspector who takes 10 ± 3 minutes to examine each one and rejects 10% of the parts. Represent the system in GPSS using the concept of facility and run the simulation for 500 parts.
3. What are the two main properties of random numbers? Test whether the 3rd, 7th, 11th, and so on numbers in the sequence in the following random number sample are auto correlated. (Z_{α}=0.05 and $Z_{0.025}$=1.96)

   0.12 0.01 0.23 0.28 0.89 0.31 0.64 0.28 0.83 0.93 0.99 0.15 0.33 0.35 0.91 0.41 0.60 0.27 0.75 0.88 0.68 0.49 0.05 0.43 0.95 0.58 0.19 0.36 0.69 0.87

Short Answer Questions: [5 marks each]

4. Explain Markov chain with suitable example.
5. What are steps involved in simulation study? Explain.
6. Generate 10 random integers using Linear congruential method where m=1000, a =19, c=6 and X_0=13.
7. Explain different estimation methods which are used in simulation output analysis.

8. Define verification and validation. Explain the process of model verification in brief.
9. What is feedback system? Explain with example.
10. What is calling population? Explain arrival and service process in a queue.
11. Explain Monte Carlo simulation method with example.
12. Write short notes on:

    a. Non-stationary Poisson process

    b. Poker test

Model:

Long Answer Questions: [10 marks each]

1. What is queuing system? Explain the elements of queuing system with example.
2. What are the properties of random numbers? The sequence of numbers 0.54, 0.73, 0.98, 0.11 and 0.68 have been generated. Use the Kolmogorov Smirnov test with level of significance (α) 0.05 to determine if the hypothesis that the numbers are uniformly distributed on the interval 0 to 1 can be rejected. (Note that the value of D for $\alpha = 0.05$ and $N = 5$ is 0.565).
3. What is a system modelling? What are the steps involved in system modelling.

Short Answer Questions: [5 marks each]

4. What are the application areas of simulation?
5. Differentiate between analog and digital computers.
6. Explain Markov chain with suitable example.
7. Why is GPSS called transaction flow-oriented language?
8. Why is confidence interval needed in the analysis of simulation output? How can we establish a confidence interval?
9. Explain Monte Carlo simulation.
10. What is three step approach for validation of simulation models?

11. Why is confidence interval needed in the analysis of simulation output? How can we establish a confidence interval?
12. Write short notes on:
    a. Non-Stationary Poisson process
    b. Feedback system

2079:

Long Answer Questions: [10 marks each]

1. What is transaction in GPSS? Explain about facility in GPSS. Customers arrive at Joey Barbershop once every 15±3 minutes and it takes Joey 18±2 minutes to cut hair of a customer. Create a GPSS model with block diagram for the Barbershop using the concept of facility and run the simulation for 9 hours.
2. Why is the accuracy of analog computers low? Explain analog computer with suitable example. Differentiate between analog and digital computers.
3. Define and develop a Poker test for four-digit random numbers. A sequence of 1,000 random numbers, each of four digits has been generated. The analysis of the numbers reveals that in 525 numbers all four digits are different, 419 contain exactly one pair of like digits, 47 contain two pairs, 9 have three digits of a kind and 7 contain all like digits. Use Poker test to determine whether these numbers are independent. (Critical value of chi-square for a = 0.05 and N = 4 is 9.49).

Short Answer Questions: [5 marks each]

4. Why Confidence interval is needed in the analysis of simulation output. How can we establish a confidence interval?
5. Explain the Monte Carlo simulation method with example.
6. Generate ten 3-digit random integers and corresponding random variables using Multiplicative Congruential method where a =7, and X0= 22.

7. “Building a model right” and “Building a right model”. Are both statements same? Discuss the importance of V&V.
8. Differentiate between discrete and continuous system.
9. What is Markov chain? Explain with example.
10. Why do we need discrete logarithm over normal logarithm? Find out whether 3 is primitive root of 7 or not.
11. Describe dynamic physical model in detail with the help of suitable example.
12. Write short notes:
    a. Hypothesis testing
    b. Stationary Poisson process

2078:

Long Answer Questions: [10 marks each]

1. Define queuing system. Explain the Kendall's notation for queuing system? What are the various performance measures in single server queuing System? Explain which of them determines system stability and how?
2. Define true random numbers and pseudo random numbers with its properties. The sequence of numbers 0.64, 0.50, 0.25, 0.58,0.72, 0.90 has been generated. Use KS Test with Da=0.050 => 0512 to determine if the hypothesis that they are uniformly distributed on interval [o, 1] can be rejected.
3. What do you understand by dynamic mathematical model? Explain with example. Differentiate it with static mathematical model.

Short Answer Questions: [5 marks each]

4. Describe the phases in simulation.
5. Explain the concept of discrete event simulation. Explain Poisson's arrival pattern.
6. Explain Monte Carlo simulation method with an example?
7. Define the terms verification, calibration, validation and accreditation of models.
8. Use Multiplicative congruential method to generate a sequence of random numbers with X=7, a=11 m=16.
9. Why are estimation methods used in simulation? Explain.
10. Explain the importance of elimination of initial bias during simulation.

11. Workers come to a supply store at the rate of one every 6(+-) 2 minutes. Their requisitions are processed by one of the two clerks who take 8 (+-) 2 minutes for each requisition. The requisitions are then passed to a single storekeeper who fills them one at a time, taking 6(+-)3 minutes for each. Draw GPSS Block diagram to simulate the above problem for 100 requisitions. Define SSL protocol. Mention the services provided by PGP.
12. Write short notes on (any two):

    a. Digital analog simulator
    b. Simulation tools

2076:

Long Answer Questions: [10 marks each]

1. Define queuing system. Explain different queuing disciplines. Also explain different performance measures for evaluation of queuing system.
2. Difference between chi-square test and KS test for uniformity. Use KS test to check for the uniformity for the input set of random numbers given below. 0.54, 0.73, 0.98, 0.11, 0.68, 0.45. Assume level of significance to be D_{α}=0.05 => 0.565.
3. What do you understand by static mathematical model? Explain with example. Differentiate between stochastic and deterministic activities.

Short Answer Questions: [5 marks each]

4. Discuss the merits and demerits of system simulation.
5. Explain Markov's chain with a suitable example.
6. Define arrival pattern. Explain non-stationary Poisson process.
7. Differentiate between validation and calibration. How can we perform validation of a model?
8. Use Mixed congruential method to generate a sequence of random numbers with X0 = 27, n = 17, m = 100 and c = 43.
9. What do you mean by replication of runs? Why is it necessary?
10. Explain generation of non-uniform random number generation using inverse method.

11. Parts are being made at the rate of one every 10 minutes. They are of two types, A and B. And are mixed randomly with about 10% being type B. A separate inspector is assigned to examine each part. Inspection of part A takes 6 ± 2 minutes. Each inspector rejects 10% of parts they inspect. Draw GPSS block diagram to simulate the above problem for 100 parts.
12. Write short notes on (any two):

    a. System and its environment
    b. Simulation run statistics.